A STACK OVER A STACK IS A STACK OVER THE BASE

THE STACKS PROJECT AUTHORS

Editorial note. No omission markers in-span; proof complete as written.

Allowed standard prerequisites (editorial).

- Fibred over fibred. categories-lemma-fibred-over-fibred (cross-chapter, named). Gives that Y is a fibred category over C, settling stack condition (1).
- Stack and descent data definitions. definition-stack, definition-descent-data, definition-effective-descent-datum (same chapter, named). Fix the three stack conditions and the shape of the descent data used in condition (3).

Source reference. Original reference 09WW: focused-seeds/geometry/S9-stk-stack-over-stack/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 09WV: focused-seeds/geometry/S9-stk-stack-over-stack/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 026B: sources/stacks-stacks/pinned-original-source.tex, source lines 498–498. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 026E: sources/stacks-stacks/pinned-original-source.tex, source lines 646–646. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original definition 026F: sources/stacks-stacks/pinned-original-source.tex, source lines 785–785. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 042V: sources/stacks-stacks/pinned-original-source.tex, source lines 388–388. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is

Date: stacks-project:stacks.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $p : \mathcal{X} \to \mathcal{C}$ be a stack. Endow $\mathcal{X}$ with the topology inherited from $\mathcal{C}$ and let $q : \mathcal{Y} \to \mathcal{X}$ be a stack. Then $\mathcal{Y}$ is a stack over $\mathcal{C}$. If p and q define stacks in groupoids, then $\mathcal{Y}$ is a stack in groupoids over $\mathcal{C}$.*

Proof. We check the three conditions in Definition 026F to prove that $\mathcal{Y}$ is a stack over $\mathcal{C}$. By Categories, Lemma 09WV we find that $\mathcal{Y}$ is a fibred category over $\mathcal{C}$. Thus condition (1) holds.

Let U be an object of $\mathcal{C}$ and let y_1, y_2 be objects of $\mathcal{Y}$ over U. Denote $x_i = q(y_i)$ in $\mathcal{X}$. Consider the map of presheaves

$$q : Mor_{\mathcal{Y}/\mathcal{C}}(y_1, y_2) \longrightarrow Mor_{\mathcal{X}/\mathcal{C}}(x_1, x_2)$$

on $\mathcal{C}/U$, see Lemma 042V. Let $\{U_i \to U\}$ be a covering and let φ_i be a section of the presheaf on the left over U_i such that φ_i and φ_j restrict to the same section over $U_i \times_U U_j$. We have to find a morphism $\varphi : y_1 \to y_2$ restricting to φ_i. Note that $q(\varphi_i) = \psi|_{U_i}$ for some morphism $\psi : x_1 \to x_2$ over U because the second presheaf is a sheaf (by assumption). Let $y_{12} \to y_2$ be the strongly $\mathcal{X}$-cartesian morphism of $\mathcal{Y}$ lying over ψ. Then φ_i corresponds to a morphism $\varphi_i' : y_1|_{U_i} \to y_{12}|_{U_i}$ over $x_1|_{U_i}$. In other words, φ_i' now define local sections of the presheaf

$$Mor_{\mathcal{Y}/\mathcal{X}}(y_1, y_{12})$$

over the members of the covering $\{x_1|_{U_i} \to x_1\}$. By assumption these glue to a unique morphism $y_1 \to y_{12}$ which composed with the given morphism $y_{12} \to y_2$ produces the desired morphism $y_1 \to y_2$.

Finally, we show that descent data are effective. Let $\{f_i : U_i \to U\}$ be a covering of $\mathcal{C}$ and let (y_i, φ_{ij}) be a descent datum relative to this covering (Definition 026B). Setting $x_i = q(y_i)$ and $\psi_{ij} = q(\varphi_{ij})$ we obtain a descent datum (x_i, ψ_{ij}) for the covering in $\mathcal{X}$. By assumption on $\mathcal{X}$ we may assume $x_i = x|_{U_i}$ and the ψ_{ij} equal to the canonical descent datum (Definition 026E). In this case $\{x|_{U_i} \to x\}$ is a covering and we can view (y_i, φ_{ij}) as a descent datum relative to this covering. By our assumption that $\mathcal{Y}$ is a stack over $\mathcal{C}$ we see that it is effective which finishes the proof of condition (3).

The final assertion follows because $\mathcal{Y}$ is a stack over $\mathcal{C}$ and is fibred in groupoids by Categories, Lemma 09WW. □